

Purna Lokesh Reddy

DevOps Engineer (AI & ML)

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SUMMARY

DevOps Engineer with **4+ years** of experience building and maintaining **cloud-native infrastructure on AWS**, specializing in **Kubernetes (EKS)**, **CI/CD automation**, and **MLOps pipelines**. Hands-on experience deploying **GPU-accelerated ML workloads** on **AWS SageMaker** and integrating **AI-driven monitoring** using Python, TensorFlow, and scikit-learn. Proven track record reducing infrastructure costs, improving deployment velocity, and maintaining **99.9%+ uptime** across high-traffic distributed systems. Skilled in **Terraform**, **ArgoCD**, **GitHub Actions**, **Prometheus**, and **Grafana** with strong DevSecOps fundamentals including **SOC 2 compliance**, **Trivy**, and **OPA policy enforcement**.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EKS, EC2, S3, RDS, Lambda, SageMaker, IAM, CloudWatch, Auto Scaling, ElastiCache)

Infrastructure as Code & Automation: Terraform, AWS CloudFormation, Ansible, Bash, Shell Scripting

Containers & Orchestration: Docker, Kubernetes (EKS), Helm, ArgoCD, RBAC, Horizontal Pod Autoscaler (HPA), Node Group Scaling, Resource Optimization

CI/CD & DevOps: GitHub Actions, Jenkins, GitLab CI/CD, AWS CodePipeline, Blue-Green Deployments, Canary Releases

MLOps & AI Infrastructure: AWS SageMaker, NVIDIA GPU Infrastructure (A10G), ML Inference Optimization, Predictive Anomaly Detection, Python (TensorFlow, scikit-learn)

Observability & SRE: Prometheus, Grafana, ELK Stack, OpenTelemetry, PagerDuty, SLI/SLO Monitoring, FinOps

Security & DevSecOps: Trivy, Open Policy Agent (OPA), AWS Secrets Manager, Kubernetes RBAC, SOC 2, PCI-DSS

Databases & Messaging: PostgreSQL (RDS), Redis (ElastiCache), MySQL, Apache Kafka, PgBouncer

Methodologies: GitOps, SRE, AIOps, Agile (Scrum/Kanban)

PROFESSIONAL EXPERIENCE

Shopify | *DevOps Engineer (AI & ML)*

Remote (USA) | August 2024 – Present

- Engineered and maintained **multi-region AWS EKS infrastructure** using **Terraform, Helm, and Ansible**, achieving **99.97% uptime** across high-traffic production environments, including Black Friday and flash-sale events.
- Optimized **GPU-based ML inference workloads** on **AWS SageMaker, Kubernetes, and NVIDIA A10G nodes**, reducing **P99 latency by 24%** for recommendation systems handling **5M+ daily requests**.
- Integrated **DevSecOps pipelines** using **GitHub Actions, Trivy, and Open Policy Agent (OPA)**, enforcing **SOC 2 compliance** and preventing non-compliant container deployments.
- Developed **AI-driven anomaly detection systems** using **Python, TensorFlow, and scikit-learn** across **200+ microservices**, reducing **MTTR by 32%** and improving incident response.
- Built **observability and FinOps dashboards** with **Prometheus, Grafana, and AWS CloudWatch**, monitoring **SLIs/SLOs, GPU utilization, and cloud spend**, identifying **\$180K+ annual savings**.
- Tuned **PostgreSQL (AWS RDS) and Redis (ElastiCache)** using **PgBouncer, read replicas, and failover automation**, reducing **query latency by 28%** with **zero downtime**.
- Automated **Kubernetes autoscaling, GitOps workflows, and infrastructure provisioning** using **ArgoCD, Helm, and Terraform**, improving **deployment reliability and release velocity**.

Uber | *DevOps Engineer*

India | August 2020 – July 2023

- Managed **AWS cloud infrastructure** and **Kubernetes (Amazon EKS)** environments supporting **300+ microservices**, ensuring **99.95% service availability** for large-scale rider and driver platforms.
- Automated **infrastructure provisioning** using **Terraform and AWS CloudFormation**, reducing **manual deployment effort by 60%** and eliminating **configuration drift** across environments.
- Led migration of **15+ legacy applications to Kubernetes**, developing **Helm charts**, configuring **Horizontal Pod Autoscaler (HPA)**, and optimizing **CPU/memory utilization**, reducing **EC2 costs by 28%**.
- Designed and optimized **CI/CD pipelines** using **Jenkins and GitLab CI/CD**, integrating **automated testing, security scanning, and deployment validation**, enabling **high-frequency production releases** with minimal rollback rates.
- Implemented **centralized observability and monitoring** using **Prometheus, Grafana, ELK Stack, and AWS CloudWatch**, improving visibility into **latency, throughput, and system health** across **50+ production services**.
- Strengthened **DevSecOps and GitOps practices** using **Trivy, ArgoCD, Kubernetes RBAC, and policy enforcement**, reducing **recurring production incidents** and improving **deployment reliability**.
- Maintained **Kubernetes RBAC, cluster upgrades, and node group scaling** for distributed systems handling **millions of daily transactions** across Uber's APAC rider and driver platforms.

EDUCATION

Master's in information technology | University of Cincinnati, OH, USA.

Graduated: Dec 12, 2024